Richard Tsai

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Experience

Cloud Application Developer 07/2022 – 08/2024

IBM

- Worked as a full-stack developer in a project with Gatsby, Apollo Client for GraphQL, Java Spring Boot and Jahia CMS
- Streamlined manual development environment setup by over 50% by implementing Docker containers and Python scripting automation
- Built automated CI/CD pipelines on Jenkins for containerized applications using Docker & Terraform Secrets Manager
- Reduced vulnerabilities by 15% through using Snyk's static code analysis, code reviews and CI/CD integration
- Sped Gatsby build time and website performance by 25% by reusing Gatsby cache in CD pipelines and optimizing business logic
- Utilized Splunk to monitor & troubleshoot various server log files and Oracle database for detection of security incidents & performance issues
- Integrated Azure Defender for Cloud into CD pipeline for real-time threat protection, vulnerability assessment, & compliance monitoring

Helpdesk Consultant 07/2019 – 05/2022

Rensselaer Polytechnic Institute

- Responded to 500+ technical incidents within campus network, resolved 90% of issues on first contact
- Communicated with students and faculty to provide support via Bomgar remote desktop and phone
- Maintained accurate records of all interactions with customers through Zendesk software
- Implemented knowledge base system, resulting in a 50% reduction in repeat calls due to similar technical issues
- Mentored new hires by training them on software applications, hardware devices, and ticketing system

Cybersecurity Defensive Operations Programming Intern 05/2021 – 08/2021

The MITRE Corporation

- Contributed as a full-stack developer in IDS/IPS in-house project with PostgreSQL and Redis, Python Flask, and VueJS
- Utilized Jira for task tracking, assignments, and collaboration with team members
- Developed custom API calls that fetched data on over 50,000 known vulnerabilities from database
- Increased code coverage by 30% through testing using pytest framework, resulting in more reliable, accurate API endpoints
- Improved mean time to detect (MTTD) for security incidents by 20%, allowing for faster response times and reduced damage

Cybersecurity Defensive Operations Intern 06/2020 – 08/2020

The MITRE Corporation

- Analyzed network PCAP files and host-based log data using Zeek and ELK stack dashboard for insights
- Automated 90% of Zeek script execution using master bash script, reducing manual effort by 75%
- Monitored and managed a web crawling application utilizing in-house Python library packages
- Implemented custom SIEM dashboard using AngularJS framework, resulting in a 40% reduction in reporting time
- Presented findings and insights to senior cybersecurity team members through bi-weekly meetings, contributed to project planning discussions

Education | Certifications

Rensselaer Polytechnic Institute 05/2022

Bachelor of Science in Computer Science

Certified Information Systems Security Professional (CISSP) 05/2028

CompTIA Security+ 08/2019

AWS Certified Security – Specialty 11/2023

Personal Projects

YACS – Open-Source Schedule Management @ RPI RCOS

Yet Another Course Scheduler – an open-source project to help students plan out their semester

- Led a full-stack project utilizing VueJS, Python FastAPI, and PostgreSQL, containerized with Docker
- Conducted weekly agile sprint meetings, used GitHub Projects to monitor ticket progress, and issued release candidates
- Implemented NGINX best practices to enhance security and maintain optimal performance, conducted CI testing
- Configured Docker containers with secure defaults, including disabling unnecessary services and setting up strict access controls
- Managed GitHub Action CI/CD pipelines and deployment via Kubernetes, ensuring high availability
- Enhanced security on staging servers on OCI, AWS, and on-premises production servers by using Terraform

Personal Home Lab

On-going personal home lab to add more future projects/servers

- Configured FortiGate NGFW with custom IPv4 policies, DoS (Denial-of-Service) policies, VPN, and network segmentation into multiple zones
- Deployed multiple on-home services across servers and containers via Kubernetes clusters with custom networking configurations
- Implemented Grafana, Prometheus and Loki to monitor server activities, network traffic, & detect potential security incidents
- Deployed custom IPS (Intrusion Prevention System) using CrowdSec, which detected & blocked 90% of all incoming threats
- Set up internal DNS servers using Unbound for High-Availability (HA) and DNSSEC validation, serving as a trust anchor for home network

Skills

Languages: Java, Python, C, C++, SQL, Bash, Node.JS frameworks, Typescript
Platforms: Windows, macOS, Redhat Linux, Microsoft Server, AWS, Azure, GCP, Docker, Kubernetes, GitHub, GitLab, BitBucket, Terraform
Tools: Wireshark, NMAP, Metasploit, Hashcat, Zeek, Grafana-Loki-Prometheus stack, ELK stack, Splunk